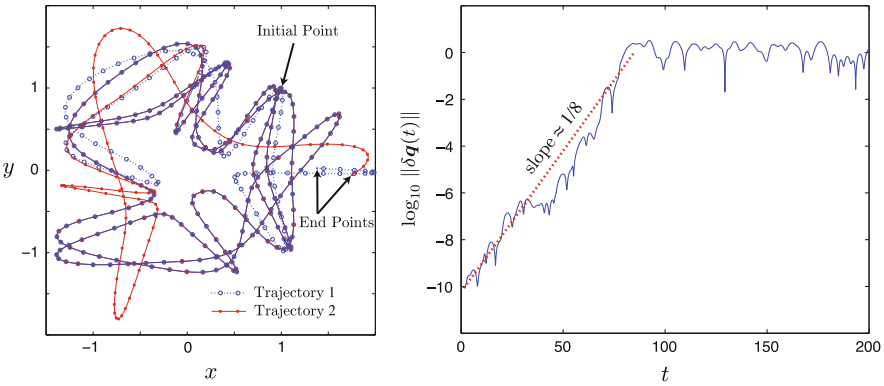


Fig. 1.25 Rapid separation of trajectories in the perturbed central force problem. The trajectories in the *left panel* are started from initial conditions differing only in the 10th digit. They are then found to separate rapidly and end up following entirely different paths. The *right panel* shows that the trajectories separate exponentially fast in the early going; eventually, once the difference is of the same order as the size of the domain, the norm of the difference stabilizes

1.7.2 Variational Equations and Lyapunov Exponents

One way of quantifying the sensitive dependence on initial conditions in a nonlinear dynamics model is via *Lyapunov exponents*. Usually this is done by introducing the *variational equations* which describe the time-dependent variation of perturbations of a solution of a dynamical system. Besides giving us a means to verify the chaotic nature of given system, the variational equations prove useful in describing the *symplectic structure* (taken up in the next chapter) which is essential to the design of effective numerical methods for molecular dynamics.

Let a dynamical system $\dot{z} = f(z)$ be given in $\mathbb{R}^m$ with flow map $\mathscr{F}_t : \mathbb{R}^m \to \mathbb{R}^m$ which we assume to be continuously differentiable. Let $z(t, \zeta)$ represent the solution of initial value problem

$$\dot{z} = f(z), \qquad z(0) = \zeta .$$

We then compute the $m \times m$ Jacobian matrix of $\mathscr{F}_t(z(t, \zeta))$:

$$W(t) = \mathscr{F}_t'(z(t, \zeta)) = \frac{\partial \mathscr{F}_t}{\partial z}(z(t, \zeta)).$$

Differentiating $W(t)$ with respect to t and using the differential equation and the chain rule, we have

$$\frac{\mathrm{d}}{\mathrm{d}t} W(t) = f'(z(t, \zeta)) W(t). \tag{1.10}$$

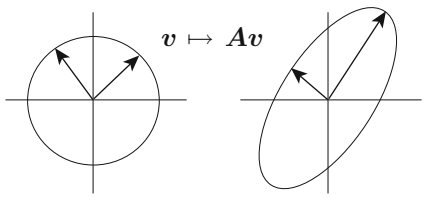


Fig. 1.26 A regular linear map takes an $m-1$-dimensional sphere to an $m-1$-dimensional ellipsoid

$\boldsymbol{W}$ satisfies a system of linear (time-dependent) differential equations involving the matrix $\boldsymbol{f}'$ (the Jacobian matrix of vector field) evaluated along the solution $\boldsymbol{z}(t)$. $\boldsymbol{W}$ may be seen as a fundamental matrix solution of the indicated linear system.

The system of Eq. (1.10) is referred to as the system of *variational equations* corresponding to the dynamical system $\mathrm{d}\boldsymbol{z}/\mathrm{d}t = \boldsymbol{f}(\boldsymbol{z})$.

If we have two solutions started from nearby initial conditions $\boldsymbol{\zeta}$, $\hat{\boldsymbol{\zeta}}$, then their difference is approximated by the solution of the variational equations

$$\boldsymbol{z}(t, \hat{\boldsymbol{\zeta}}) - \boldsymbol{z}(t, \boldsymbol{\zeta}) \approx \boldsymbol{W}(t)(\hat{\boldsymbol{\zeta}} - \boldsymbol{\zeta}).$$

The usefulness of the variational equations lies, however, in the fact that they are scale independent, one may view them as defining the relative variation induced over time of an infinitesimal perturbation of the initial condition.

We may view a regular linear mapping $\boldsymbol{v} \mapsto \boldsymbol{A}\boldsymbol{v}$, where $\boldsymbol{A} \in \mathbb{R}^{m\times m}$, as a mapping of an $m-1$-dimensional sphere (embedded in the m-dimensional Euclidean space) to an ellipsoid in the same space (Fig. 1.26). The square roots of the eigenvalues of $\boldsymbol{A}^T\boldsymbol{A}$, also called the singular values of $\boldsymbol{A}$, then give the axes of the image ellipsoid.

In the case of the transformation $\boldsymbol{v} \mapsto \boldsymbol{W}(t)\boldsymbol{v}$, the singular values of the transformation give insight into how much the ellipse is expanded or contracted in different directions over time. The Lyapunov exponents $\lambda_1, \lambda_2, \ldots, \lambda_m$ are defined by

$$\lambda_i = \limsup_{t\to\infty} \frac{1}{t} \log \sigma_i(\boldsymbol{W}),$$

where σ_i represents the ith singular value of the given matrix (to maintain continuity, these should be ordered in some way, say largest to smallest). The lim sup is needed as we wish to ignore any initial fluctuations and are only interested in the long-term behavior. The presence of a positive Lyapunov exponent implies exponential growth of perturbations, which, as we have seen, is one of the hallmarks of chaos.

This is not a very easy computation to carry out in the general case. The difficulty lies in the fact that the limit taken in the definition may require a very long simulation, typically millions or billions of time-steps, and the eigenvalues must be computed at each step. Even for modest-sized systems a sophisticated approach is needed [146, 343]. We have used special purpose software [105], resulting in an estimate of the largest Lyapunov exponent of approximately 0.28. Observe that in the right hand panel of Fig. 1.25, the growth of the logarithm of the trajectory